

Verification of whether final matrix is in echelon form

1. There are no zero rows in the matrix. $\therefore$ The condition that the non-zero rows must be above the zero rows is vacuously True.
2. Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 3 in Column 2
Row 3 Leading Entry = 3 in Column 4
 $\therefore$ The condition that the leading entry of each row must be in a column to the right of the column of the leading entries of all rows above it, is True.
3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
 $\therefore$ The condition that each pivot column has only 0's below the pivot position is True.

The matrix $\begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ is in echelon form.

There exists a row in the matrix of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. By Theorem 2, the system is inconsistent, i.e. no solution set exists.

$\therefore b = \begin{bmatrix} 3 \\ -7 \\ -3 \end{bmatrix}$ is not a linear combination of the vectors formed from the columns of the matrix $A = \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 5 \\ -2 & 8 & -4 \end{bmatrix}$ $\square$

$$14. A = \begin{bmatrix} 1 & -2 & -6 \\ 0 & 3 & 7 \\ 1 & -2 & 5 \end{bmatrix}, b = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$$

Solution: We need to check if for some $x_1, x_2, x_3 \in \mathbb{R}$,
$$x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$$

This is equivalent to checking whether the augmented matrix $\begin{bmatrix} 1 & -2 & -6 & 11 \\ 0 & 3 & 7 & -5 \\ 1 & -2 & 5 & 9 \end{bmatrix}$ has a solution set.